Beau Dougan

San Diego, California
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Education

California State University San Marcos **Graduated May 2026 Cum Laude**

Bachelor of Science in Software Engineering

Relevant Coursework: Data Structures and Algorithms (**C++ & Java**), Operating Systems (**C++**), Software Testing (**Junit**), Embedded Systems (**Assembly & C**), Discrete Math, Software Engineering (**Java**), Database Systems (**SQL**), Software Architecture (**Java**), Computer Architecture, Machine Learning (**Python**)

Technical Skills:

- Languages: **Java**, **Assembly, C++, Python**, **SQL, Embedded C++**
- Tools/Technology: Git, VS Code, Unix, Junit, IntelliJ, Eclipse, SQL, AWS (Intermediate), Jenkins, scikit-learn

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Projects:

Qualcomm Robotics Senior Design Project September 2025 – May 2026

- Collaborated within a four-person **Agile**/**SCRUM** team to build an autonomous voice-activated guide robot using Python and **ROS2**, implementing **LiDAR** and **IMU** odometry using **EKF** for navigation and **BLE** for iOS device communication.
- Produced a robot able to move around a pregenerated 3D map created using **SLAM Toolbox** while guiding the user to their destination.
- Engineered requirements, software integration, and testing, producing comprehensive technical documentation across the full engineering lifecycle.
- Made a tough decision to switch from monocular SLAM using a **Camera Pi 3** and **IMU odometry** to **LiDAR odometry** and SLAM modelling trading off affordability for accuracy and performance
- Met weekly with a Qualcomm project manager throughout the rapid prototyping process.

STM32 Door Access Simulator 2026

- Worked in a team of three to create a keypad-based access control system on an STM32 Nucleo board using Embedded C++, integrating PWM servo control, keypad scanning, digital I/O, and embedded state management to simulate door lock and unlock behavior with LED and buzzer feedback.
- Due to a weak servo motor, we made a decision to simulate a lock instead of having the door be forced closed by the servo as originally planned.

Credit Card Fraud Detection 2026

- Created a machine learning model which can detect fraud with 96% recall using anonymized features from credit card purchases using **pandas** and **scikit-learn**
- Used **scikit-learn** to build random forest and logistic regression models as well as comparing the two models

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Additional Work Experience:

Doordash Delivery Driver	February 2024 - 2026
Mainstream Source Office Assistant	August 2022 - 2024
Sambazon Açaí Café Barista	August 2020 – 2022
Thai Society Delivery Driver	August 2017 – 2020
In-N-Out Burger	July 2014 – 2017